Heng Xu

Backend & AI Engineering Portfolio

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Master of Computer Science (Data Science and AI), University of Sydney, Jul 2026. I turn ambiguous requirements into testable Python services, reproducible model evaluation, governed workflows and handover-ready technical evidence.

Python / FastAPI | AI systems & evaluation | API contracts | CI / testing | Available on one week's notice

01 AI Hub PII Redaction Service - USYD Capstone (High Distinction)

Mar 2026 - Jun 2026 | 8-person client-delivered project | FastAPI, Pydantic, pytest, Docker, GitHub Actions

From text redaction to a testable multi-format service

- Built core FastAPI/Pydantic services, offset-based redaction, Block/Review policy loading, versioned API contracts and pytest coverage.
- Extended ingestion across TXT, DOCX, PDF and image inputs with OCR/parser metadata and stable structured span output.
- Established lint/test/image-build checks and validated a client deployment on an on-prem DGX Spark through Tailscale with API and operations handover documentation.
- Contributed to fine-tuning/distillation and led vLLM-based comparison of teacher, base-student and fine-tuned-student models.

14,000	2,000	0.9775	0.9904
evaluation examples	iteration subset	overall F1	high-risk F1



DOCX upload through `/v1/redact-file` with structured entity spans and masked output.



Image/OCR path returning aligned spans for Australian PII entity types.

> Evidence boundary: client deployment validation and reproducible handover; no claim of public production traffic or independent ownership of the full team project.

02 Governed Knowledge & Workflow Tool (LLM-KB)

Mar 2026 - May 2026 | Personal engineering project | Private repository; walkthrough available

A governed knowledge workflow for reliable AI-agent use - not a generic RAG demo

905 governed source documents → Policy-gated intake & upgrade → SQLite truth + YAML / JSON Schema → Deterministic products + doctor / review / MCP

- Separated ontology truth from product data and generated outputs, with deterministic regeneration and integrity checks.
- Used policy gates for intake and upgrades, plus append-only audit manifests and approval/rejection evidence.
- Exposed CLI/MCP interfaces while keeping mutation paths controlled and reviewable.
- Validated schemas, references, synchronisation, environment, policy, freshness, review queues and pipeline health through repository doctor checks.

Engineering Evidence

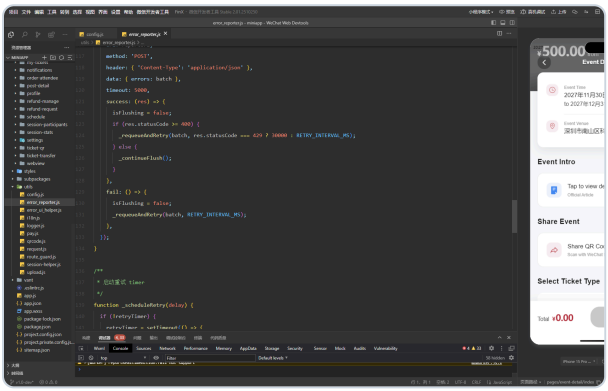
Email | GitHub | LinkedIn

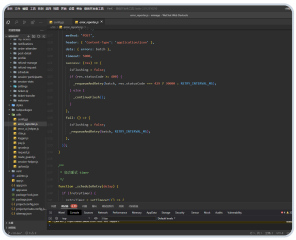
Application delivery, data workflows and local AI systems

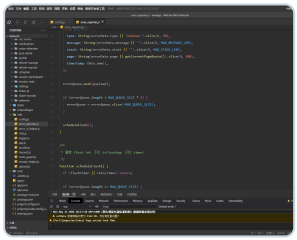
Selected supporting work showing integration checks, QA/UAT evidence, deterministic data processing, service orchestration and operational handover.

03 FinX Application Delivery & Financial Data Automation

Dec 2025 - Jun 2026 | Finfo Global internship | Django/DRF integration, QA/UAT, Python data workflows







Turning requirements into observable delivery evidence

- Supported Django/DRF API integration, QA/UAT and handover by converting requirements into API checks, acceptance criteria, regression evidence and release gates.
- Covered registration, ticketing, payment preparation, check-in, refund/transfer, content and IM workflows without claiming production ownership.
- Independently built Python workflows over company-provided datasets from 11 banks: cleaning, field mapping, return/volatility/drawdown/fee metrics, R1/R2/R3 scoring, Excel delivery and rapid recalculation as rules changed.

> The screenshots are work-sample evidence from the development environment. They demonstrate workflow coverage and role-dependent surfaces, not public production usage.

04 Luna - Local Desktop AI Assistant

Nov 2025 - Dec 2025 | Open-source personal project | Electron, FastAPI, llama.cpp, local/cloud routing

Repeatable local AI runtime instead of a single-model demo

- Designed an Electron + FastAPI assistant with llama.cpp-served Qwen2 models and dual-model routing.
- Implemented local/cloud fallback, repeatable startup orchestration, health checks and configuration validation.
- Kept private runtime data portable through isolated user-data roots and local-first routing controls.

Electron | FastAPI | Qwen2 | llama.cpp | Health checks

